SMART INDIA HACKATHON 2026

Problem Statement ID - 26231

Problem Statement Title -Digital Companion for Field Drug Testing

Theme - MedTech / BioTech / HealthTech

PS Category - Software

Team ID - K26234

Team Name - HexTech



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PROPOSED SOLUTION

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Problem

India: 1,20,010 NDPS cases registered in 2023 (NCRB, Crime in India 2023)

- Officers read the kit's colour change by eye, so street lamp, torch or sunlight and personal judgement can change the result.
- There is no standard colour value to compare the result with.
- The test often leaves no clear digital record of who tested, where and when, so it is hard to verify later.

Solution

- **Check the Test** - the app photographs the kit next to a ₹10 colour card, corrects the colour and says POSITIVE, NEGATIVE or NOT SURE.
- **Record the Test** - saves the photo fingerprint, internet time, GPS and officer ID as a signed record with a QR check.
- **Manage the Records** - supervisors see the test log, map, NOT SURE review and PDF reports on the dashboard.

User Experience

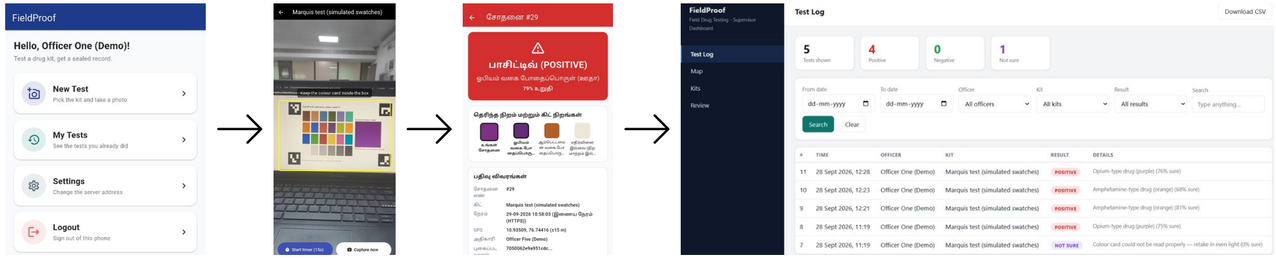
- **4 steps** - choose kit, start the kit's read timer, take the photo inside the box, see the result. About 16 s per test (19 scans, 1 team member).
- **NOT SURE path** - the app says why (blurry, shiny, card not seen, colour unclear) and asks for a retake. Still NOT SURE means no field result. The supervisor can only order a retest or confirm NOT SURE, and the sample goes to FSL.
- **Easy to read** - big coloured result, works in Tamil too.

Prototype Demo Video: Prototype Video Link

Our Approach

Test Kit → Take Photo → Check Colour → Get Result → Save Proof → Admin Review

Prototype Flow - real screenshots of our working prototype



Real screenshots. Result screen shown in Tamil. Test Log (right) filters by date, officer, kit and result, and downloads CSV or PDF with QR.

Novelty

Not just a colour reader. A field result that anyone can check later.

	Eye check today	FSL lab	FieldProof
Result on the spot	Yes	No, report later	Yes, about 16 s
Same reading in any light	No	Yes, lab conditions	Yes, colour card in every photo
Says NOT SURE when unclear	Depends on officer	Not needed	Yes, with the reason
Record anyone can check later	No	Lab report	Signed record + QR, BSA s.63 page
Confirms which drug	No	Yes	No, first check only

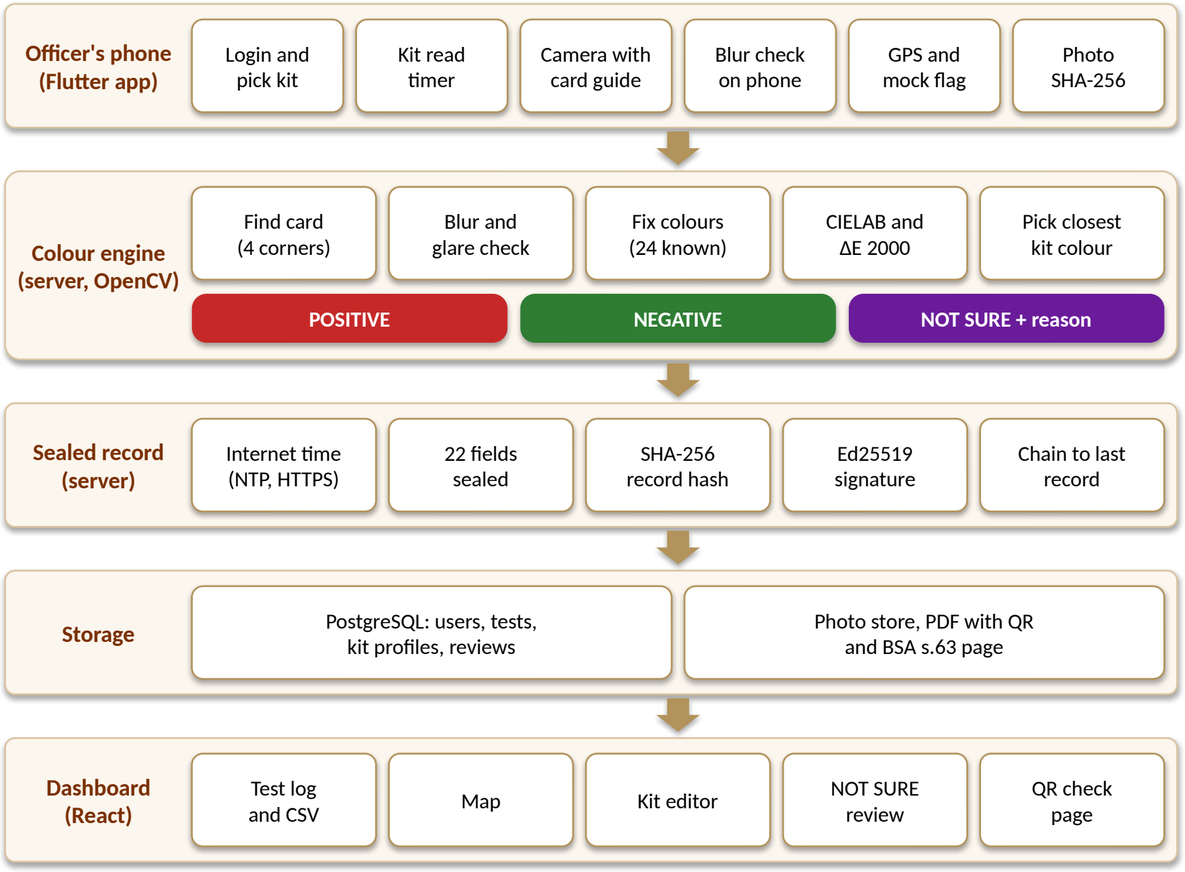
Also: new kit profiles are added from the admin website without an app update. The app works in Tamil.

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TECHNICAL APPROACH

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System Architecture



Everything in this diagram is built and running in our prototype.

How the App Decides

- The card has 24 colours we already know. The app sees how the light changed them and removes the same change from the kit colour.
- Result only if the closest kit colour is within **ΔE 14** and at least **4 points** ahead of the next. Else NOT SURE.
- **Card fit check** - after correction, the 24 known colours must be within ΔE 9 on average (real photos gave 5.9 to 8.7).
- The % on the result screen is a **match score** (closeness and lead), not a probability.
- **No AI model, on purpose** - it needs no drug photos to train, and every result can be explained with numbers.

Measured Results

0 wrong answers so far, and every NOT SURE had a reason.

Test	n	What happened
Computer-made photos	37	5 lights. 25 of 25 good photos right, 12 of 12 bad photos NOT SURE
Real scans, 28 Sep	6	5 right, 1 NOT SURE (card fit), 0 wrong
Real scans, 29 Sep	19	9 right, 7 of 7 trick cards NOT SURE, 3 blurry photos NOT SURE, 0 wrong
All real scans	**25**	**14 right, 11 NOT SURE, 0 wrong**
Speed	18	Server check 2.5 s typical, 5 s max Whole test about 16 s (1 team member)

Kit colours are printed swatches shown on a screen, 1 phone, 1 room, our team only. 0 wrong in 14 real answers is a small test, not proof. Real kits need an FSL trial (lab prototype, TRL 4).

Technology Stack

- Flutter (Dart)
- FastAPI (Python)
- OpenCV + NumPy
- CIELAB + ΔE2000
- PostgreSQL + SQLAlchemy
- React + Vite
- JWT + bcrypt
- SHA-256
- Ed25519
- NTP
- Leaflet + OpenStreetMap
- ReportLab

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FEASIBILITY AND VIABILITY

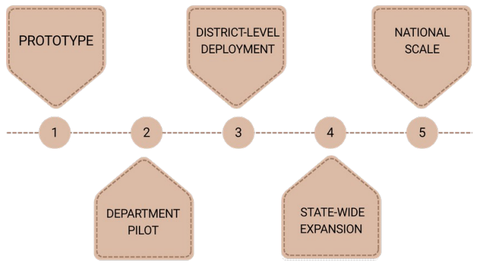
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Feasibility & Sustainability

Legal: first check only, the FSL lab confirms it.
Privacy: no face, fingerprint or suspect name, only a sample code.

- **Technical** - the full chain works today: phone app, server, database and dashboard (lab prototype, TRL 4).
- **User-friendly** - one photo, result in about 16 s. Tested by our team, not yet by officers.
- **Low cost** - officer's own Android phone, a ₹10 reusable card per officer and open-source software. Kits cost the same as today.
- **Scalable** - new kits are added from the admin website, after FSL calibrates their colours with real reagent.
- **Sustainable** - state police IT cell runs it, FSL keeps kit colours correct (proposed).

Phased Implementation & Scalability



Pilot Plan (proposed)

- **Where** - 1 district in Tamil Nadu, 20 officers, 3 months, with the nearest FSL.
- **Check 2 things apart** - app vs an expert reading the same kit (target 95%+ same), and kit vs FSL lab (kit error, not app error).
- **Size** - 60+ tests per kit, so 0 wrong would show error under 5%.
- **Pass if** - NOT SURE under 10% and every record passes the QR check.
- **Cost** - about ₹9,200 for cards and server only (estimate). Kits, phones, FSL time, training and audit are extra.

Potential Challenges & Strategies

Real-world problem	What our app does	Status
Bad light (street lamp, torch)	24 known colours on the card correct the kit colour	Built
Blurry or shiny photo, card missing	NOT SURE with the reason and a retake. Blur check on the phone too	Built
Colour in between two results	NOT SURE. Supervisor can only order a retest or confirm NOT SURE, sample goes to FSL	Built
Kit colour keeps changing	Each kit has a read time, the app timer takes the photo on time	Built
Fake GPS app	Android mock-location flag is sealed in the record	Flag only
Photo of a screen or a staged colour	Screen-photo check and Play Integrity phone check	Planned
Someone with the server key rewrites records	Changes by others are caught. Next: key in a hardware box (HSM), chain copy kept outside	Planned

Cost & Deployment Plan

After the pilot, the cost can come from the state police or NCB IT budget. Server about ₹36,000 a year (estimate).

COST	HOSTING	MAINTENANCE	DEPLOYMENT	SCALE
Own phone, ₹10 card per officer, open-source software	NIC cloud or state data centre in India (to confirm)	Security update every 3 months. FSL updates kit colours	Pilot first, then district, state and national	New kits from the website after FSL calibration

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IMPACT AND BENEFITS

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Officer Benefits

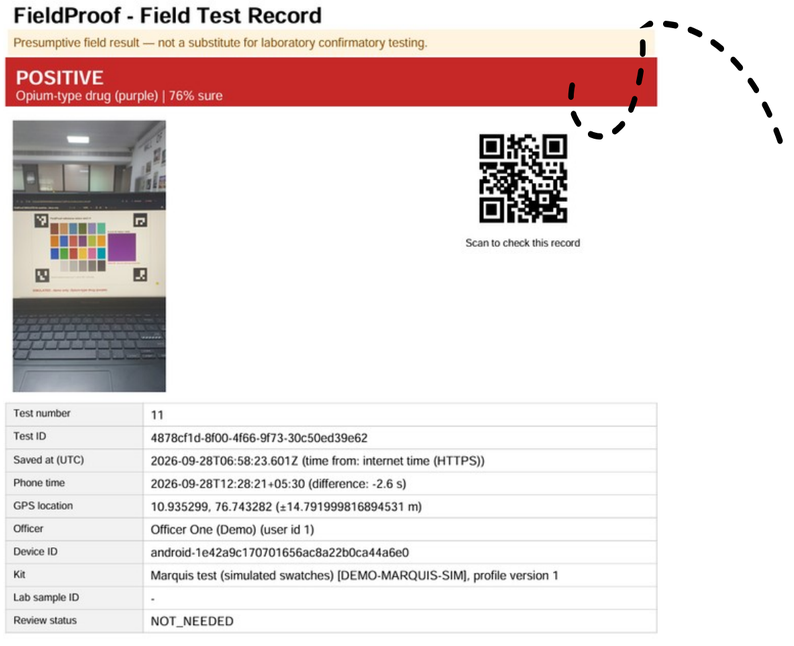
- Same reading in any light
- NOT SURE instead of a guess
- Result in about 16 s
- Record made automatically
- Works in Tamil

Organisation Benefits

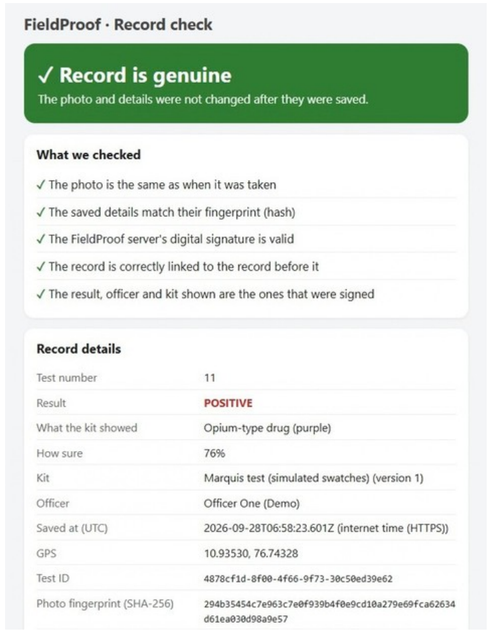
- Time, place, officer and photo for every test
- Any change to a record is caught
- PDF with QR and BSA s.63 page
- Search, map and CSV export
- NOT SURE tests go to supervisor review

QR-Based Record Verification

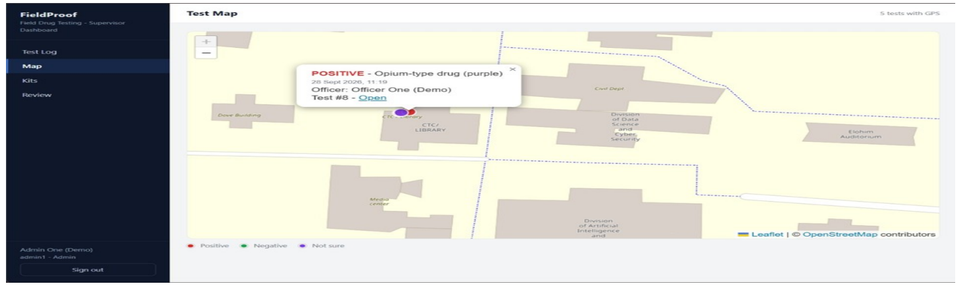
Judges can scan it on their own phone. For the demo the QR opens a public HTTPS check page (verify only, nothing else is open).



Test number	11
Test ID	4878cf1d-8f00-4f66-9f73-30c50ed39e62
Saved at (UTC)	2026-09-28T06:58:23.601Z (time from: internet time (HTTPS))
Phone time	2026-09-28T12:28:21+05:30 (difference: -2.6 s)
GPS location	10.935299, 76.743282 (±14.791999816894531 m)
Officer	Officer One (Demo) (user id 1)
Device ID	android-1e42a9c170701656ac8a22b0ca44a6e0
Kit	Marquis test (simulated swatches) [DEMO-MARQUIS-SIM], profile version 1
Lab sample ID	-
Review status	NOT_NEEDED



Map - Geo-Tagged test result



Court Certificate: every PDF report now has a filled BSA section 63 page (photo hash, phone, time, GPS) for the officer and expert to sign. Open a full sample report (PDF, 2 pages)

Tamper Test - we tried it on our saved records

Test #10 `prev 2ece5c80` `hash 6b1b4e7f` → **Test #11** `prev 6b1b4e7f` `hash 03dd9f0c` → **Test #12** `prev 03dd9f0c` `hash a110cff6`

Ed25519 signature of test 11
`JRIdu6ERgadC+2n6chKfUStp...`
Photo SHA-256
`294b35454c7e963c7e0f939b...`

What we changed (test #11)	Check page	Failed check
Nothing (original)	GENUINE	none, all 5 passed
1 byte of the photo	TAMPERED	photo
Result inside sealed copy	TAMPERED	hash, signature, fields
Result in database row	TAMPERED	fields
The signature	TAMPERED	signature

We tried this on our real saved records on 29 Sep 2026, only in memory. Each test keeps the hash of the test before it.

Future Scope (our plan after SIH)

- Month 1-2 - offline mode and phone security chip signing, built together.
- Next - FSL trial with real kits, paper card on 3+ phones, officer testing.
- Month 3 - screen-photo check, Play Integrity, Hindi and voice (Tamil done).
- After pilot - apply to NIC or MHA for e-Sakshya and CCTNS (not contacted yet).

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RESEARCH AND REFERENCES

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What they found	How we use it	Research
IEEE EMBC 2016 Phone camera colours change with light and phone model, and reference colours can help handle this.	Our ₹10 colour card fixes colours in every test	[1] H. Karisen and T. Dong, "Illumination and device independence for colorimetric detection of urinary biomarkers with smartphone," in Proc. 38th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC), Orlando, FL, USA, 2016, pp. 5184-5187, doi: 10.1109/EMBC.2016.7591895.
IEEE TPS-ISA 2024 Digital evidence must maintain integrity and show how it was handled.	Photo fingerprint + signature + chained log	[2] S. Nath, K. Summers, J. Baek, and G.-J. Ahn, "Digital evidence chain of custody: Navigating new realities of digital forensics," in Proc. IEEE 6th Int. Conf. Trust, Privacy Secur. Intell. Syst. Appl. (TPS-ISA), Washington, DC, USA, 2024, pp. 11-20, doi: 10.1109/TPS-ISA62245.2024.00012.
Forensic Science International 2014 A mobile phone camera can measure drug-test colours using RGB values.	Our app reads the test colour as a number	[3] A. Choodum, K. Parabun, N. Klawach, N. Nic Daeid, P. Kanatharana, and W. Wongniramaikul, "Real time quantitative colourimetric test for methamphetamine detection using digital and mobile phone technology," Forensic Sci. Int., vol. 235, pp. 8-13, Feb. 2014, doi: 10.1016/j.forsciint.2013.11.018.
Analytical Chemistry 2016 Reading the colour by eye is subjective, and phone image analysis can give an objective reading.	App gives POSITIVE / NEGATIVE / NOT SURE	[4] S. T. Krauss et al., "Objective method for presumptive field-testing of illicit drug possession using centrifugal microdevices and smartphone analysis," Anal. Chem., vol. 88, no. 17, pp. 8689-8697, Sep. 2016, doi: 10.1021/acs.analchem.6b01982.
Digital Discovery 2023 Computer vision can quantify and compare colour changes that may look similar by eye.	App shows photo colour next to kit colours	[5] N. Bugeja et al., "Teaching old presumptive tests new digital tricks with computer vision for forensic applications," Digital Discovery, vol. 2, no. 4, pp. 1143-1151, 2023, doi: 10.1039/D3DD00066D.

What we add: earlier work shows a phone can read test colours [1][3][4][5]. We add the NOT SURE rule, a signed record anyone can check with a QR, and a BSA section 63 page.

Corrected colour. Honest NOT SURE. A record anyone can check.

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TEST RESULTS IN DETAIL

Backup slide for judges' questions

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All Tests in One Table

Test	n	Setup	Result
Computer-made photos	37	5 lights: day, bulb, tube light, dim, street lamp. Rules were tuned on these	25/25 good right, 12/12 bad NOT SURE
Real scans, 28 Sep	6	Card shown on a phone or laptop screen	5 right, 1 NOT SURE, 0 wrong
Real scans, 29 Sep	19	8 positive, 4 plain, 7 trick cards. 1 room, window light	9 right, 10 NOT SURE, 0 wrong
All real scans	**25**	1 phone (Nothing A015), our team only	**14 right, 11 NOT SURE, 0 wrong**
Stress test	171	19 real photos x 9 computer-made changes (dark, yellow bulb, tilt, shake, grain)	70 right, 101 NOT SURE, 0 wrong. Sharp photos 69/72 right
Tamper test	5	Copies of saved records, 4 changes and 1 untouched	4/4 changes caught, original passed
Speed	18	Server, 6 real photos x 3 runs	2.5 s typical, 5 s max

What 0 wrong means

0 wrong in 14 real answers is a small test. By the rule of three the true error rate could still be up to about 21%.

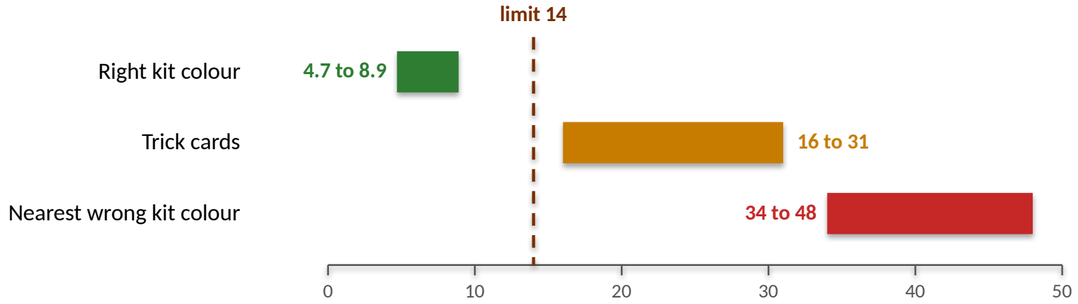
To show under 5% we need 60+ tests with 0 wrong, so the pilot asks for 60+ tests per kit.

Photo 7 on 28 Sep was rejected when the card fit limit was lower. We quote that live result, not a re-run.

What we do not claim

- Accuracy on real reagents (not tested yet)
- Which drug it is (the FSL lab confirms)
- Court approval. It supports the BSA s.63 certificate, the officer and expert still sign
- Results on other phones or with officers (next step)

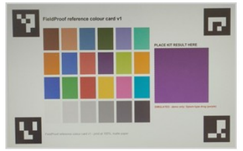
Colour Gap on Real Scans (ΔE 2000)



Right colours sat well under 14 and wrong kit colours far above it. The closest trick card was only about 2 points over the limit, so each real kit gets its own limit from FSL data.

Card fit (average error left on the 24 known colours after correction): 5.9 to 8.7 on real photos of a screen, limit 9. Under 1 on computer-made photos. Paper card test is next.

Why the 29 Sep Scans Said NOT SURE


Purple card
POSITIVE 79%


Orange card
POSITIVE 76%


Plain card
NEGATIVE 75%


In-between card
NOT SURE


Green card (not in kit)
NOT SURE


Blurry photo
NOT SURE

- **3 positive cards** - photo was blurry (records 13, 14, 20). The app asked for a retake.
- **7 trick cards** - 3 matched no kit colour, 4 were stopped earlier by photo checks (1 blurry, 2 card not found, 1 card fit).
- Stress test: all 38 NOT SURE on real cards came from blurry photos.

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SECURITY AND FSL TRIAL

Backup slide for judges' questions

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Threat Model (what is built, what is open)

Threat	What stops it today	Next step	Status
Photo or record changed later	Photo SHA-256 on phone and server, signed record, chain, QR check	Supervisor counter-seal, officer DSC	Built, 4/4 caught
Photo of a screen or staged colour	In-app camera only, no gallery upload	Screen-photo (moire) check, Play Integrity	Open
Fake GPS app	Android mock-location flag sealed in the record	Play Integrity device check	Partly
Wrong phone clock	Server takes internet time, phone time and the gap are also saved	Outside time-stamp (RFC 3161)	Built
Insider with the server key	Key only on the server, public key is published	Key in a hardware box (HSM), chain head copied outside daily	Open
Server key file lost	Server makes a new key, old records then fail the check	Stop and alert instead, keep a key backup	Fix planned
Public QR page shows officer name and exact GPS	Page is verify only, nothing else is open	Hide the name, round off GPS	Fix planned
Test done with no network	Not saved today	Offline queue built together with phone chip signing	Planned
One chain for a whole state	One chain today (prototype)	Chain per device or a daily batch root	Planned

Legal hooks: BSA 2023 s.63 (certificate for electronic records, with hash), BNSS 2023 s.105 (recording of search and seizure), NDPS SSSD Rules 2022, DPDP Act 2023 s.8(5) (keep data safe).

FSL Trial to Set Each Kit's Limit

1. **Real reagent**
   FSL runs each kit on known samples, 3+ kit lots, photo at the kit's read time and again later to see colour drift.

2. **Phones and light**
   Paper card, 3+ phones, daylight, yellow bulb, tube light and street lamp.

3. **Measure**
   Card fit on paper, spread of the right colour, distance to the nearest wrong colour.

4. **Set limits**
   ΔE limit and lead for each kit from that gap, with a NOT SURE band in between. Freeze them before testing.

5. **Test fresh samples**
   Samples not used for tuning, compared with the FSL lab. Report agreement with a range, not one number.

New kit, step by step

FSL calibrates the kit, admin adds the kit profile (it gets a version number), a small field check, then officers can use it. Kit lot and expiry will be saved in each test record (planned).